

CONJUGATION AND EXCESS IN THE STEENROD ALGEBRA

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ABSTRACT. In this paper we prove a formula involving the canonical antiautomorphism χ of the mod-2 Steenrod algebra $\mathcal{A}(2)$, namely,

$$\begin{aligned} \chi(Sq^{2^j(2^{i+1}-1)}Sq^{2^{j-1}(2^{i+1}-1)}\dots Sq^{(2^{i+1}-1)}) \\ = Sq^{2^i(2^{j+1}-1)}Sq^{2^{i-1}(2^{j+1}-1)}\dots Sq^{(2^{j+1}-1)}, \end{aligned}$$

and discuss its implications for the study of the image of the $\mathcal{A}(2)$ -action on $\mathbb{F}_2[x_1, \dots, x_s]$.

1. INTRODUCTION

1.1. Notation. The Milnor basis of the mod-2 Steenrod algebra is indexed by sequences $R = (r_1, r_2, \dots)$ of nonnegative integers almost all of which are 0 [M]. We denote the corresponding basis element by $\langle R \rangle$; its dimension is $|\langle R \rangle| = \sum_i (2^i - 1)r_i$. If $R = (r, 0, 0, \dots)$, then the corresponding basis element, abbreviated $\langle r \rangle$, is the Steenrod square Sq^r . The element $\langle a_1 \rangle \cdots \langle a_n \rangle$ is *admissible* if $a_r \geq 2a_{r+1}$ for $r < n$ and $a_n > 0$ if $n > 1$. The admissible elements form an additive basis of $\mathcal{A}(2)$ [SE], but though the objects of study in this paper are certain admissible elements, the calculations are best expressed in terms of the Milnor basis. In particular, define $\mathcal{B}(\theta)$ for $\theta \in \mathcal{A}(2)$ to be the set of Milnor basis elements appearing in θ , so that $\theta = \sum \{ \langle R \rangle : \langle R \rangle \in \mathcal{B}(\theta) \}$.

The Steenrod algebra acts on $\mathbb{F}_2[x_1, \dots, x_s]$, the polynomial algebra on elements x_i of dimension 1, which is the mod-2 cohomology ring of

$$\overbrace{\mathbb{R}P^\infty \times \dots \times \mathbb{R}P^\infty}^s.$$

Following Singer [Si], we say that a polynomial F is *hit* if it is the image of the positively graded part of $\mathcal{A}(2)$, that is, if $F = \sum_{i>0} Sq^i F_i$ for some polynomials F_i [Pe, W]. The *excess* of an element $\theta \in \mathcal{A}(2)$ is given by $\text{ex}(\theta) = \min\{\hat{s} : \theta(x_1 x_2 \cdots x_{\hat{s}}) \neq 0 \in \mathbb{F}_2[x_1, \dots, x_s]\}$ (cf. [K]). Since linear maps commute with the action of $\mathcal{A}(2)$, it follows that whatever s might be, $\theta(F) = 0$ for any polynomial in $\mathbb{F}_2[x_1, \dots, x_s]$ of degree $< \text{ex}(\theta)$. One has that $\text{ex}(\langle n \rangle) = n$

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and, more generally, that $\text{ex}(\langle a_1 \rangle \cdots \langle a_n \rangle) = \sum_{r=1}^{n-1} (a_r - 2a_{r+1}) + a_n$ if $\langle a_1 \rangle \cdots \langle a_n \rangle$ is admissible [SE].

Given a pair (k, a) of nonnegative integers, we define $T(k; a) = \langle 2^k a \rangle \cdots \langle 2a \rangle \langle a \rangle$. Note that $|T(k; a)| = (2^{k+1} - 1)a$, that $T(k; a)$ is admissible of excess a , and that on polynomials of degree $|A| = a$ one has $T(k; a)(A) = A^{2^{k+1}}$.

We denote by χ the canonical antiautomorphism of $\mathcal{A}(2)$.

1.2. Results. In [D] Davis computes $\chi T(j; 1)$:

Theorem 1.1 [D]. *For all $j \geq 0$, we have $\chi T(j; 1) = \langle 2^{j+1} - 1 \rangle$.*

Our Theorem 3.1 is a generalization of Theorem 1.1.

Theorem 3.1. *For all integers i and j , we have $\chi T(j; 2^{i+1} - 1) = T(i; 2^{j+1} - 1)$.*

Davis's argument proves also that $\chi T(j; 2) = \langle 2(2^{j+1} - 1) \rangle$, leading one to conjecture the following.

Conjecture 1.2. $\chi T(j; 2(2^{i+1} - 1)) = T(i; 2(2^{j+1} - 1))$ for all $j \geq 0$.

Bruner, using a computer, has verified that this conjecture holds for pairs i, j such that $i \leq 4$ and $|T(i; 2(2^{j+1} - 1))| \leq 255$.

The pattern of these formulas for $\chi T(j; 2^k)$, $k = 0, 1$, breaks down for $k = 2$; it is not true that $\chi T(1; 4) = \langle 12 \rangle$.

Theorem 3.1 and Conjecture 1.2, if it is true, allow us to identify new families of hit monomials, as we discuss in §4.

2. TECHNICAL LEMMA

We begin by recalling a proposition from the proof of Theorem 1.1. For integers $a = \sum a_i 2^i$ and $b = \sum b_i 2^i$, where $a_i, b_i \in \{0, 1\}$, we say a dominates b if $a_i \geq b_i$ for all i , and write $a \succeq b$.

Proposition 2.1 [D]. $\langle m \rangle \chi \langle n \rangle = \sum \{ \langle R \rangle : |R| = m + n; |R| + \sum r_i \succeq 2m \}$.

An argument similar to Davis's describes the result when the order of factors is reversed.

Proposition 2.2. $\chi \langle n \rangle \cdot \langle m \rangle = \sum \{ \langle R \rangle : |R| = m + n; \sum r_i \succeq m \}$.

Note that in both cases, the question of whether an $\langle R \rangle$ of the relevant degree appears as a summand depends only on $\sum r_i$. The formulas are very similar, and the following lemma, used repeatedly in the inductive proof of Theorem 3.1, combines them to give a way of pushing a factor involving χ through a product. Specifically, it permits us, in certain cases, to write $\langle m \rangle \chi \langle n \rangle$ as the sum of $\chi \langle p \rangle \cdot \langle q \rangle$ and two terms of the form $\langle m' \rangle \chi \langle n' \rangle$, which with luck will be more tractable than the original $\langle m \rangle \chi \langle n \rangle$.

Lemma 2.3. *Suppose that k, l, m , and n are nonnegative integers such that:*

- (1) $k > l$,
- (2) $m + n = 2^k - 2^l$,
- (3) $m < 2^{k-1}$,
- (4) $m \equiv 0 \pmod{2^l}$.

Then

$$\begin{aligned} Sq(m) \cdot \chi Sq(n) &= \chi Sq(n - m - 2^l) \cdot Sq(2m + 2^l) + Sq(m + 2^{k-1} + 2^{l-1}) \\ &\quad \cdot \chi Sq(n - 2^{k-1} - 2^{l-1}) + Sq\left(\frac{m+n}{2}\right) \cdot \chi Sq\left(\frac{m+n}{2}\right), \end{aligned}$$

with the understanding that $\chi Sq(s) = 0$ if $s < 0$, and that the rightmost two terms vanish when $l = 0$.

Proof. For simplicity, we will refer to the i th term in the equation above as Q_i . To prove the lemma, it is enough to show that $\mathcal{B}(Q_1) = \mathcal{B}(Q_2) \amalg \mathcal{B}(Q_3) \amalg \mathcal{B}(Q_4)$, where $X \amalg Y$ is the symmetric difference of the two sets. For $Sq(T) = Sq(t_1, t_2, \dots)$ a basis element of degree $|Sq(T)| = 2^k K - 2^l$, write $L_T = 2^k K - 2^l + \sum t_i$ and $R_T = \sum t_i$. By Propositions 2.1 and 2.2, $Sq(T) \in \mathcal{B}(Q_1)$ (resp. $\mathcal{B}(Q_2)$, resp. $\mathcal{B}(Q_3)$, resp. $\mathcal{B}(Q_4)$) $\Leftrightarrow L_T \succeq 2m$ (resp. $R_T \succeq 2m + 2^l$, resp. $L_T \succeq 2^k + 2m + 2^l$, resp. $L_t \succeq 2^k - 2^l$). By (3) and (4) above, $\mathcal{B}(Q_3)$ and $\mathcal{B}(Q_4)$ are $\subseteq \mathcal{B}(Q_1)$. Writing $L_T = (R_T - 2^l) + 2^k K$ and using (3) and (4), one finds that also $\mathcal{B}(Q_2) \subseteq \mathcal{B}(Q_1)$. Similarly, expressing R_T in terms of L_T , one may check that $\mathcal{B}(Q_1) \subseteq \mathcal{B}(Q_2) \amalg \mathcal{B}(Q_3) \amalg \mathcal{B}(Q_4)$. Observe that if $Sq(T) \in \mathcal{B}(Q_1)$, then $Sq(T) \in \mathcal{B}(Q_4) \Leftrightarrow \sum t_i < 2^l$. Finally, note that $\mathcal{B}(Q_1) = \mathcal{B}(Q_2)$ when $l = 0$. This proves the lemma. $\square$

3. PROOF OF THE THEOREM

We are now ready to prove the main result of this paper.

Theorem 3.1. *For all integers i and j , we have $\chi T(j; 2^{i+1} - 1) = T(i; 2^{i+1} - 1)$.*

Proof. Clearly it suffices to prove the theorem under the assumption that $i \leq j$. Theorem 2.1 gives the result for $i = 0$ and all j , so we proceed inductively by assuming that $\chi T(\hat{i}; 2^{j+1} - 1) = T(\hat{j}; 2^{i+1} - 1)$ for $\hat{i} \leq i - 1$ and all $\hat{j}$ and for $\hat{i} = i$ and all $\hat{j} \leq j - 1$. The inductive proof will draw on the following remark: under the above assumptions,

$$(1) \quad \begin{aligned} \chi Sq(2^{l-1}K) \cdot T(l-1; 2^{i+1} - 1) &= 0 \quad \text{for all } K \equiv 1 \pmod{2} \\ &\text{and all } 1 \leq l \leq i. \end{aligned}$$

For we have

$$\begin{aligned} \chi Sq(2^{l-1}K) \cdot T(l-1; 2^{i+1} - 1) &= \chi[\chi T(l-1; 2^{i+1} - 1) \cdot Sq(2^{l-1}K)] \\ &= \chi[T(i; 2^l - 1) \cdot Sq(2^{l-1}K)] \\ &= \chi[T(i-1; 2(2^l - 1))Sq(2^l - 1) \cdot Sq(2^{l-1}K)], \end{aligned}$$

and it follows from the Adem relations that $Sq(2^l - 1)Sq(2^{l-1}K) = 0$. This verifies (1).

Now consider

$$(2) \quad \begin{aligned} \chi T(i; 2^{j+1} - 1) &= \chi[T(i-1; 2^{j+1} - 1)] \cdot \chi\langle 2^i(2^{j+1} - 1) \rangle \\ &\stackrel{\text{ind}}{=} T(j; 2^i - 1) \cdot \chi\langle 2^i(2^{j+1} - 1) \rangle. \end{aligned}$$

Beginning from the right, we multiply $\chi\langle 2^i(2^{j+1} - 1) \rangle$ by the successive terms $\langle 2^l(2^i - 1) \rangle$, $0 \leq l \leq j$, of $T(j; 2^i - 1)$. At the cost of an error term, Lemma 2.3 allows us to push the factor involving χ past the l th singleton, transforming

both in the process. Furthermore, (1) ensures that the error term vanishes upon multiplication by the $(l-1)$ -singletons to its right. Namely, we prove that

$$(3) \quad \begin{aligned} & T(l; 2^i - 1) \cdot \chi(2^i(2^{j+1} - 1)) \\ &= \chi(2^i(2^{j+1} - 2^{l+1})) \cdot T(l; 2^{i+1} - 1) \quad \text{for } 0 \leq l \leq j. \end{aligned}$$

The case $l = 0$ follows from Lemma 2.3. Suppose then that the claim holds for $l - 1$. Then, as $T(l; 2^i - 1) = \langle 2^l(2^i - 1) \rangle \cdot T(l - 1; 2^i - 1)$, we have

$$\begin{aligned} & T(l; 2^i - 1) \cdot \chi Sq(2^i(2^{j+1} - 1)) \\ &= Sq(2^l(2^i - 1)) \cdot \chi Sq(2^i(2^{j+1} - 2^l)) \cdot T(l - 1; 2^{i+1} - 1) \quad [\text{induction}] \\ &= [\chi Sq(2^i(2^{j+1} - 2^{l+1})) \cdot Sq(2^l(2^{i+1} - 1))] \cdot T(l - 1; 2^{i+1} - 1) \quad [\text{Lemma 2.3}] \\ &\quad + [Sq(2^{i+j} + 2^{i+l} - 2^{l-1}) \cdot \chi Sq(2^{i+j} - 2^{i+l} - 2^{l-1})] \cdot T(l - 1; 2^{i+1} - 1) \\ &\quad + [Sq(2^{i+j} - 2^{l-1}) \cdot \chi Sq(2^{i+j} - 2^{l-1})] \cdot T(l - 1; 2^{i+1} - 1). \end{aligned}$$

But the second and third summands vanish by (1), so we have

$$\begin{aligned} & T(l; 2^i - 1) \cdot \chi Sq(2^i(2^{j+1} - 1)) \\ &= \chi Sq(2^i(2^{j+1} - 2^{l-1})) \cdot [Sq(2^l(2^{i+1} - 1)) T(l - 1; 2^{i+1} - 1)] \\ &= \chi Sq(2^i(2^{j+1} - 2^{l-1})) \cdot T(l; 2^{i+1} - 1). \end{aligned}$$

This establishes (3). Finally, taking $l = j$, we find that

$$\begin{aligned} \chi T(i; 2^{j+1} - 1) &= T(j; 2^i - 1) \cdot \chi(2^i(2^{j+1} - 1)) \\ &= \chi \langle 0 \rangle \cdot T(j; 2^{i+1} - 1). \end{aligned}$$

This proves the theorem. $\square$

4. HIT ELEMENTS

In [K] Kraines gives a proof that the excess $\text{ex}(\chi\langle n \rangle)$ is given by $\mu(n)$, where $\mu(n)$ is the number of summands in the most efficient way of writing n as a sum of numbers of the form $2^i - 1$; that is, $\mu(n) = \min\{m : n = \sum_{i=1}^n (2^{k_i} - 1)\}$ for some integers k_i . The following generalization follows immediately from Theorem 3.1:

Corollary 4.1. $\text{ex}(\chi T(i; 2^{j+1} - 1)) = 2^{i+1} - 1 \quad (= (2^{i+1} - 1)\mu(2^{j+1} - 1)).$

In view of its consequences in the study of hit monomials, described below, Corollary 4.1 leads to

Conjecture 4.2. $\text{ex}(\chi T(i; a)) = (2^{i+1} - 1)\mu(a).$

The case $a = 2$, i arbitrary, would follow from Conjecture 1.2. Using a computer, Bruner has verified Conjecture 4.2 for all pairs (i, a) such that $a \leq 31$ and $|T(i; a)| \leq 255$.

The connection of Conjecture 4.2 with hit monomials is as follows: Recall that $\alpha(n)$ denotes the number of 1's in the binary expansion of the integer n . In [W] Wood extends and proves a conjecture due to Peterson.

Theorem 4.3 [W]. *Let M be a monomial of degree $|M| = d$, and suppose that e of its exponents are odd. If $\alpha(d + e) > e$, then M is hit.*

The proof involves writing M in the form $EF^2 = E \cdot \langle f \rangle(F)$, where E is squarefree of degree e and $f = |F|$, and showing that modulo hit elements, $E \cdot \theta(F) \equiv \chi\theta(E) \cdot F$ for any $\theta \in \mathcal{A}(2)$. The result then follows from the fact that $\text{ex}(\chi\langle f \rangle) = \mu(f)$; one must check that the assumption $\alpha(d + e) > e$ implies that $\mu(f) > e$.

Singer has conjectured a generalization of Theorem 4.3, using a condition which involves not the degree of the squarefree part of M but rather the degree of the 2^{k+1} -powerfree part. That is, write $M = EF^{2^{k+1}}$ where E contains no 2^{k+1} -powers, and let $e = |E|$, $f = |F|$. The conjecture can be paraphrased in part as follows.

Conjecture 4.4 [Si]. *Suppose that M is a monomial with decomposition $M = EF^{2^{k+1}}$ as above. If $e < (2^{k+1} - 1)\mu(f)$, then M is hit.*

This time, we have that

$$EF^{2^{k+1}} = E \cdot [\langle 2^k f \rangle \cdots \langle 2f \rangle \langle f \rangle(F)] = E \cdot [T(k; f)(F)],$$

which modulo hit elements is congruent to $[\chi T(k; f)(E)] \cdot F$; but this element vanishes if $\text{ex}(\chi T(k; f)) > |E|$, so we see that Conjecture 4.2 implies Conjecture 4.4. In the cases for which Conjecture 4.2 has been verified, we can state the following.

Theorem 4.5. *Suppose that M is a monomial with decomposition $EF^{2^{k+1}}$ as above and that $e < (2^{k+1} - 1)\mu(f)$. If $f = 2$ or $f = 2^i - 1$ for some i , or if (k, f) satisfies $f \leq 31$ and $(2^{k+1} - 1)f < 255$, then M is hit.*

In unpublished work, Singer has verified Conjecture 4.4 for $k = 1$ and all f using different techniques.

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